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Design and validation of an instrument of self-perception regarding the lecturers' use of ICT resources: to teach, evaluate and research

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Abstract

This study is part of a research about the use of ICT resources by Higher Education teachers to teach, evaluate and research (UICT-TER). The aim, in this case, is the creation and validation of a measuring instrument on the use of ICT resources based on the UICT-TER model. After the preparation and application of the instrument to a sample of 867 Higher Education teachers in the Spanish territory, the reliability, content validity, comprehension and construct validity were analyzed. In the initial creation of the instrument, three dimensions corresponding to a total of 24 items were established, achieving a final version of 16 items. Regarding content validity, the experts ($n = 4$) positively assessed the construct, dimensions and items of the questionnaire. The reliability analysis through Cronbach's alpha were highly satisfactory and in relation to the Confirmatory Factor Analysis, the results found a good fit of the model. In addition, the instrument had external evidence of validity through different descriptive analyses with the variables gender, type of university (public-private), employment relationship (full-part-time), area of knowledge of the teaching staff (Science and Engineering-Architecture, Health Sciences, Art-Humanities and Social-Legal Sciences). Taking into account all the results obtained, the instrument created meets the psychometric properties required to be considered a valid and reliable instrument to measure the use of ICT resources according to the UICT-TER model in the Higher Education.

Keywords Lecturers · Higher education · Information and communication technologies (ICT) · Digital competence · Questionnaire

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1 Introduction

One of the main responsibilities of higher education institutions is providing a competent professional qualification to future school-leavers (Fernández-Márquez et al. 2018) in order to address the needs of the European Higher Education Area (EHEA) (Salajan 2018): people who will be able to select, update, and use knowledge in a specific context, as well as being able to learn how to learn during their lives (Bozu and Canto 2009). In this sense, the implementation of a university system based on competences it requires a deep reflection from teaching staff regarding how to systematise their students' teaching and learning processes (Biggs 2005; Hernández 2017) to support them in their research and innovation, since both elements must be part of a whole – that is, their professional profile builds on their teaching and research role (Hernández 2009; Cunha 2015; Lazaro 2018) – without forgetting that evaluation provides an integrated function in teaching and learning processes (Bozu and Canto 2009). In this systematisation process, information and communication technologies (ICT) plays a key role (European Commission 2012) since they contribute to the development of quality pedagogical and research processes (Dellit 2001), due to the fact that teacher-researchers must act as builders and creators of knowledge (Hernández 2009) in that they must produce science, research and search (Castro et al. 2004). Thus, and due to the great proliferation of technological devices and applications both in the educational field and outside it, it is essential that university teachers possess adequate digital pedagogical competence (Gudmundsdottir and Hatlevik 2018; Helleve et al. 2019; Kelentrić et al. 2018). This must be understood in a transversal way and configured in three dimensions: technological knowledge, attitudes towards ICT, and educational use (From 2017; Rivera-Laylle et al. 2017). However, it should be noted that this study is primarily interested in the use dimension (Guillén-Gámez et al. 2019), specifically the use of ICT in the teaching, evaluation, and research of university teachers.

It must be borne in mind that these teachers are currently undergoing a constant evaluation of their teaching and research activity, but in most cases, both functions are independently evaluated, so evaluative function has been left aside, for this reason, this work aims to systematise the three functions, teaching, evaluation and research. When carrying out a bibliographic review of the literature concerning the university teachers' digital competence in how they teach, evaluate, and carry out research with regard to ICT, studies have largely focused on the analysis of teaching activity (Díaz 2016; Lagoa-Varela et al. 2018) or on the research activity (Buela-Casal et al. 2017; Pérez and Aldás 2018; De-Moya-Anegón et al. 2019), while other studies have focused on how university teachers integrate ICT into their teaching practice (Lawrence and Tar 2018; Tondeur et al. 2018; Sánchez-Mena et al. 2018). However, no research could be identified that analyses the evaluative activity of teachers, nor any studies where the didactic use made by university professors was jointly contemplated taking into account different ICT resources to teach and evaluate, as well as to carry out research.

Measuring ICT skills has been a challenge among researchers in recent years, since in order to carry out research of this nature, it is necessary to rely on valid and reliable instruments that assess the said level of acquisition, appropriate to the population that is the object of study. Studies analysing the psychometric properties of digital competence have mainly focused on students in the primary and secondary education stages (Calvo et al. 2017; Martínez-Serrano 2018), university students (Flores-Lueg and Vila 2016;

Sánchez et al. 2019; Suwanroj et al. 2019; Almeraya and Tobón 2019), future education teachers (Gutiérrez Castillo et al. 2017; Cabezas González et al. 2017; Tondeur et al. 2017; Lázaro-Cantabrana et al. 2019), or active teachers (Trejo 2017; Nordén et al. 2017; Ghomi and Redecker 2018). However, the scientific literature regarding these kinds of studies focused on university teachers, is rather scarce; moreover, no instruments have been developed that analyse psychometric properties regarding the use of ICT resources in relation to teaching, evaluation, and research; as such, this can be considered the main strength of the present work, providing a broader orientation to previous studies. A summary of the main instruments used in prior research is given in the table below (Table 1), where their corresponding psychometric properties are analysed, as well as the dimensions considered in their evaluation.

As shown in Table 1, not all the studies displayed here considered the validity (exploratory and confirmatory) of instruments, and none of them carry out invariance analysis of gender in their respective constructs. Thus, they are mainly focused on content or construct validity through exploratory analysis. However, the present research considers reliability using different techniques, as well as content, comprehension, and construct validity and invariance by gender, which demonstrates an innovation in regard to the validity of the tool developed.

For this reason, the aim of this study was to design an instrument that evaluates the perception of Higher Education teachers in relation to the use they make of ICT resources for teaching, evaluating their students, and university research (UTIC Model-EEI). In addition, to demonstrate that it is an adequate instrument, its psychometric properties have been analysed. This study was carried out with methodological rigour with respect to determining the validity and reliability of the instrument, which supports the measurement questionnaire. Furthermore, this study is in accordance with the article by Guillén-Gámez et al. (2020) who used this UTIC-EEI Model to analyse the use of higher education teachers according to gender and the branch of knowledge to which they belong.

2 Investigation methodology

2.1 Proposed instrument

The creation of the UTIC-EEI instrument was carried out drawing on a wide and comprehensive perspective, based on the written literature related to the teachers' perceptions regarding their use of ICT resources when teaching and evaluating students, as well as performing task research. The instrument with the final version of the measurement, together with the specifications in the form of qualitative labels, is presented in Appendix 1.

2.2 Application of the instrument to a sample

The researchers held continuous face-to-face and virtual meetings for several months to justify the creation of the instrument and to define the subject to be evaluated, the people who will be evaluated, and the final purpose. To achieve this, we considered the need, innovation, relevance and viability of the research project. After a thorough

Table 1 Studies that analyse the psychometric properties of measurement instruments

Title of manuscript	Sample, items, reliability and validity	Dimensions and their reliability
Towards the design of a diagnostic instrument for university "technological competences of teachers" Cabero et al. (2010)	▪Sample: 70 university teachers ▪Items: 70 ▪Total reliability: .976 ▪Validity: content (judgement of 7 experts)	1. Technical (AT). $\alpha = .935$ 2. Pedagogical (AP). $\alpha = .927$ 3. Social, ethical, and legal (AS). $\alpha = .854$ 4. School management and organisation (AGE). $\alpha = .831$ 5. Communication with new tools (AC). $\alpha = .921$ 6. Professional development (ADP). $\alpha = .846$ 7. Related to the application of ICT at the University of Seville (USE). $\alpha = .851$
Digital Competence Questionnaire of the university teaching staff of the Spanish Faculties of Educational Sciences Agreda et al. (2017)	▪Sample: 1145 university teachers ▪Items: 112 ▪Total reliability: .920 ▪Validity: content (judgement of 6 experts) and construct (Exploratory Factor Analysis)	1. Use and technological literacy. $\alpha = .958$ 2. Educational methodology through ICT in the classroom. $\alpha = .953$ 3. Training of university teachers in ICT. $\alpha = .964$ 4. Attitude towards ICT in higher education. $\alpha = .826$
SABER-ICT questionnaire: level of use and appropriation of ICT in higher education teachers Taquez et al. (2017)	▪Sample: 15 university teachers ▪Items: 25 ▪Total reliability: not specified ▪Validity: content (judgement of 8 experts)	1. Technological dimension. $\alpha =$ does not specify
Instrument for the certification of university teachers in knowledge and use in information and communication technologies (ICT) and learning and knowledge technologies (TAC). González Jaimes et al. (2019)	▪Sample: 36 university teachers ▪Items: 43 ▪Total reliability: .78 ▪Validity: content (judgement of 36 experts), and construct (exploratory)	1. General considerations. $\alpha = .77$ 2. Academic activities. $\alpha = .82$ 3. Use of educational tools. $\alpha = .75$
Certification of Tic competence of university teachers. Durán et al. (2016)	▪Sample: not specified ▪Items: 44 ▪Total reliability: not specified ▪Validity: content (number of judges not specified)	1. Knowledge bases that support action with ICT. $\alpha =$ does not specify 2. Design, implementation, and evaluation of the action with ICT. $\alpha =$ does not specify 3. Individual and/or collective critical reflection of the action with ICT. $\alpha =$ does not specify

Table 1 (continued)

Title of manuscript	Sample, items, reliability and validity	Dimensions and their reliability
Preparation and validation of a questionnaire to evaluate the educational use of the Moodle e-learning platform in the university context Medina-Rebollo et al. (2019)	▪Sample: 77 university teachers ▪Items: 90 ▪Total reliability: not specified ▪Validity: content (judgement of 77 experts)	1. Knowledge and training of teachers on the Moodle platform. α = does not specify 2. Didactic use of the Moodle platform. α = does not specify 3. Advantages of using the Moodle platform. α = does not specify 4. Shortcomings and limitations of the Moodle platform. α = does not specify
Comparative Study on Perception and use of ICT among Professors of Public and Private Universities Tapasco and Giraldo (2017)	▪Sample: 147 university teachers ▪Items: 23 ▪Total reliability: .774 ▪Validity: construct (exploratory)	1. Online activities with students. α = .751 2. Satisfaction with the institution's technological resources. α = .860 3. Emerging interaction tools. α = .694 4. ICT skills. α = .761 5. Class tools. α = .659 6. Perceived utility of ICT. α = .693 7. Technostress. α = N.A.
The training of university teachers and students in the use of ICT and Moodle Correa and Rivadulla-López (2017)	▪Sample: 46 university teachers ▪Items: 43 ▪Total reliability: not specified ▪Validity: content (judgement of experts, number not specified).	1. Digital knowledge. α = .874 2. Digital skills. α = .868 3. ICT training. α = .950 4. Training in Moodle. α = .889
Development of a validated instrument to measure ICT teachers' competences for university education in the digital age. Ge et al. (2018)	▪Sample: 1147 university teachers ▪Items: 28 ▪Total reliability: .967 ▪Validity: content (judgement of experts, number not specified) and construct (exploratory and confirmatory)	1. Consciousness. α = .95 2. Literacy. α = .88 3. Strategy. α = .93 4. Research. α = .83

review of the literature on how scientific competence has been measured and, specifically, the didactic use of ICT resources, the authors proposed a number of dimensions and their respective items. The process of creating and validating the instrument was carried out following the recommendations proposed by Carretero-Dios and Pérez (2005). The procedure followed is shown in Fig. 1.

The data was collected through surveys of higher education teachers in the Spanish territory (Table 2 shows the technical data sheet for sampling). The study population consisted of 120,383 higher education teachers from the Spanish Educational System (MECD, 2018– 2019). The fieldwork was carried out between January and February 2019. However, the sample was selected intentionally, first by collecting the email addresses of 12,538 teachers through the websites of their relevant educational institutions; and later, by contacting all of these teachers by the same means, respecting the confidentiality and privacy of their contributions. The response rate was 1206 teachers, which subsequently was reduced to 867 participants as a consequence of a thorough selection of the database. Hair et al. (2009) stated that, as a general rule, it is necessary to obtain a number of observations between five and ten times greater than the number of items. In the sample of this study (which consisted of 867 teachers, and a measurement instrument consisting of 16 items) a ratio of 54:19 was obtained, far exceeding the recommended amount.

In terms of professional profile, 46.14% ($N = 400$) of the sample belonged to the branch of knowledge of Social and Legal Sciences; 21.11% ($N = 183$) to Sciences and Engineering-Architecture; 19.95% ($N = 173$) to Health Sciences; and 12.80% ($N = 111$) to the Art and Humanities branch. Regarding the sociodemographic characteristics of the teaching staff, the sample consisted of 415 women (47.9%) and 452 men (52.1%). Concerning the age range, teachers aged less than 39 years comprised 22.6% of the sample, while 54% of the subjects were aged between 40 and 49 years; 23.4% of teachers were 50 years old or older.

Regarding the (maximum) accreditation obtained by a public body, for example ANECA (National Agency for Quality Assessment and Accreditation of Spain), 23.9% ($N = 207$) of the sample did not possess any type of accreditation; 11.9% ($N = 103$) of the subjects had the category of Assistant Doctor; 20.9% ($N = 181$) had the category of Contracted Doctor; 29.3% ($N = 254$) had the category of Permanent Teacher; and finally, 14.1% ($N = 122$) of the teaching staff had the category of Professor.

2.3 Preparation of the questionnaire and method used

To develop the instrument, we designed a system of dimensions and items supported by the contributions of relevant researchers including Cabero et al. (2010), Durán et al. (2016), Tapasco and Giraldo (2017), Agreda et al. (2017), Taquez et al. (2017), Tondeur et al. (2017), Tourón et al. (2018), and Ge et al. (2018) among others, whose contributions were adapted to the context of this study.

In order to validate the developed instrument, we analysed its reliability and validity. To determine validity – that is, the degree to which the instrument measures the variable it is intended to measure, we examined the validity of content, comprehension, and construct.

To test the content validity, four experts in the field participated in three rounds of evaluation. In the first round, the construction of the instrument and its dimensions

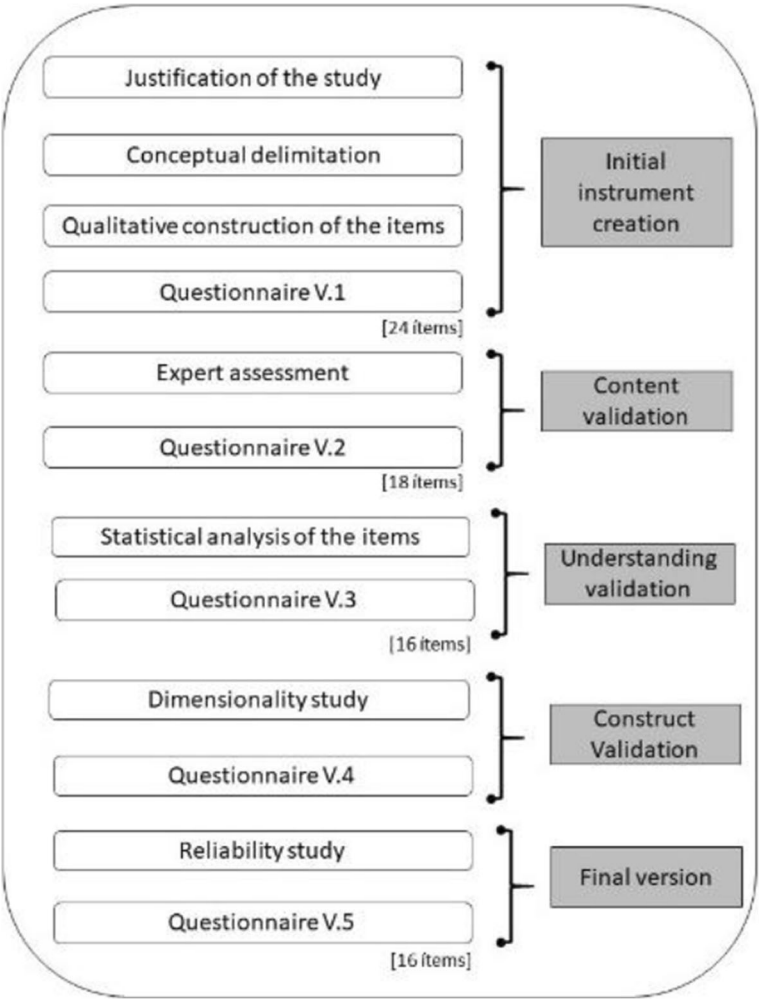


Fig. 1 Representation of the validation process of the UTIC-EEI instrument

were evaluated; in the second round, the items were evaluated (adequacy and relevance for each dimension of the construct, and adequacy in writing); and finally, according to

Table 2 Technical data of the present study

Characteristics	Survey
Universe	Teaching staff in higher education throughout the Spanish territory
Sampling procedure	Non-probabilistic, intentional
Information gathering	Personal survey to teachers via email
Sample size	867 subjects
Fieldwork	May 2018–February 2019

the recommendation of Cortada (Cortada de Kohan 1999), we eliminated items that did not exceed an average of 4 for both clarity and relevance in terms of the degree of agreement among experts.

The degree of understanding of the items was then analysed. For this, we calculated the discrimination level of the scores of each item and its corresponding dimension, using the corrected correlation coefficient. Those items with a value greater than 0.40 were considered adequate for inclusion in the final instrument.

In order to analyse the internal structure of the instrument (construct validity), the sample of participants was randomly divided into two sub-samples. The first was used to perform exploratory factor analysis (EFA) using the maximum likelihood method with Oblimin oblique rotations; this method is the most widely used in scientific works, as its results are usually adequate in most situations (Kline 2005; Fan et al. 1999). The second sub-sample was used to study the structure of the dataset through confirmatory factor analysis (CFA) and, therefore, to evaluate the data as fitting a certain theoretical factor structure (Thompson 2004; Henson and Roberts 2006). Data analysis was performed using the IBM SPSS v22 and AMOS v22 statistical programs.

To check the CFA, the dimensionality of the instrument was analysed by applying the Kaiser-Meyer Olkin (KMO) test to measure sampling adequacy, and the Barlett sphericity test. To perform the CFA, the following values were calculated: chi-square ratio on the degrees of freedom ($\chi 2$/g.l; CMIN/DF) taking into account that values less than 3 indicate a good fit (Bentler 1989); the comparative fix index (CFI) and the incremental fix index (IFI, .955), considering values higher than .90 to indicate a good fit (Hu and Bentler 1999); and the root mean square error of approximation (MSE, .058),where values below 0.06 indicate a good fit of the model.

External validity. Descriptive statistics were calculated to determine the significance of the predictive value in the three dimensions and for different study variables. Thus, factorial ANOVA was used to evaluate the variable 'area of knowledge to which the teaching staff belong' (Science and Engineering-Architecture, Health Sciences, Art-Humanities, and Social-Legal Sciences), as t-Student, to analyse and internally compare the variable 'gender' (male, female), type of university (public, private), and employment relationship (full- or part-time). The level of statistical significance was $p < .05$.

3 Results

3.1 Comprehension validity

The results of the scale discrimination indices and corrected total element correlation of the items were appropriate, being higher than 0.30 (Table 3). Those items with lower values were removed from the instrument for subsequent analyses. Furthermore, column related to reliability of the total instrument through Cronbach's alpha if each of the items is eliminated, is also excellent with values close to 0.89 for each.

3.2 Construct validity

The results determined the acceptance of the sample through the KMO Index and Bartlett's sphericity criterion (Table 4).

Table 3 Analysis of the scale discrimination index

	Scale mean if item has been deleted	Scale variance if item has been deleted	Total item correlation corrected	Cronbach's Alpha if item has been deleted
D1. Use of ICT resources to teach				
Item 1	50.6353	155.510	.543	.883
Item 2	50.4676	157.007	.548	.883
Item 3	51.4183	148.710	.573	.881
Item 4	50.7740	154.117	.497	.884
Item 5	51.2125	148.029	.593	.880
D2. Use of ICT resources to evaluate				
Item 6	51.0828	149.193	.581	.880
Item 7	51.3781	155.164	.394	.888
Item 8	51.0425	154.135	.522	.883
Item 9	51.2796	149.067	.626	.879
D2. Use of ICT resources to research				
Item 10	52.0336	147.346	.567	.881
Item 11	52.3870	148.341	.534	.882
Item 12	52.2461	149.428	.521	.883
Item 13	51.5078	149.035	.512	.883
Item 14	52.3087	148.263	.511	.884
Item 15	51.6868	145.543	.596	.880
Item 16	52.1633	144.460	.642	.878
Item 17	50.2145	143.143	.396	.840
Item 18	50.1323	143.045	.342	.831

The EFA results revealed that the 16 items were grouped under correct theoretical dimension they should remain, with a rotation conversion of 18 iterations (Table 5). The Oblimin rotation revealed the presence of three factors which explained 59.89% of the instrument's true variance. Thus, the total variance across the three dimensions was explained by three self-values, which indicates that the instrument has a high construct validity – that is, the variables included in the questionnaire reflect the true theoretical meaning of what it is intended to measure.

Table 6 shows the three identified factors and the factor importance of each of their corresponding items. Items 13, 15, 14, 10, 11, 12, and 16 were consistent with factor 1; items 6, 7, 8, and 9 with factor 2; and items 1, 2, 3, 4, and 5 with factor 3. Factor 3 refers to the didactic use by university teachers of ICT resources to teach in their area of knowledge. Factor 2 represents the didactic use by teachers of ICT resources to evaluate students in their relevant subjects. Factor 1 indicates the use by teachers of ICT resources to research and publish scientific articles. Regarding the values obtained for each of the items, the minimum value was .407 and the maximum was .861. Only two items (10 and 3) fell under two dimensions, which indicates that the items adequately relate to factors they intend to measure, so they are well designed.

Table 4 KMO Index and Bartlett's sphericity criterion (λ = 1)

KMO	Bartlett's criterion		
	X^2	gl	p
.916	3244.474	120	.001

Table 7 presents the matrix of correlations between factors, which shows that it would not have been more appropriate to apply the method of orthogonal rotations (varimax), since the factors are correlated, which points to the uni-dimensionality of the instrument consisting of a three-dimensional base.

Once the EFA was complete, a CFA was carried out. Figure 2 shows the proposed model where the standardised regression coefficients (factor loads) between the three dimensions of the instrument and their corresponding items have high values (Uriel and Aldás 2005). These relationships are positive and significant, indicating a strong relationship between the dimensions and the items comprising them. This relationship strength in a CFA is indicative of a robust model. The CFA showed the adequacy of the goodness of fit index (Fig. 2): CMIN/DF = 2.51, p = <.001; CFI = .957; TLI = .947; IFI = .957; RMSEA = .059 (90% CI = .050–.068). Therefore, the proposed model in CFA confirms the factor structure formulated in the EFA, consisting of three correlated latent variables, which again confirms that the presented instrument has good internal consistency, since it measures what it claims to measure.

3.3 Reliability analysis

The overall reliability of the instrument was determined using Cronbach's Alpha, with a very satisfactory value (α = .89). In addition, the reliability of each of the dimensions of the instrument was also calculated using the Cronbach, Spearman-Brown, and Guttman alpha coefficients (Table 8). All the coefficients produced very satisfactory results. In addition, the Omega McDonald coefficient was calculated, indicating the excellent reliability of the three dimensions together (ω = .91).

Finally, the correlations between the dimensions of the instrument were analysed using Pearson's coefficient. Table 9 shows that the correlations between the different dimensions are all positive and significant. The highest correlations occur between dimensions 1 and 2 (r = .639), and between dimensions 1 and 3 (r = .414), with the

Table 5 Total variance explained (λ = 1)

Component	Initial self-values		
	Total	% of variance	% accumulated
1	6.14	38.40	38.40
2	2.39	14.94	53.33
3	1.05	6.56	59.89

Table 6 Exploratory factor analysis of the UTIC-EEI instrument

	Factors		
	1	2	3
Item 13	.861		
Item 15	.843		
Item 14	.743		
Item 10	.675	.302	
Item 11	.640		
Item 12	.585		
Item 16	.486		
Item 6		.666	
Item 7		.643	
Item 9		.641	
Item 8		.551	
Item 4			.692
Item 2			.605
Item 5			.557
Item 3		.323	.472
Item 1			.407

lowest coefficient between dimensions 2 and 3 (r = .399), thus indicating that this instrument is reliable – in other words, that it has little measurement error.

3.4 Invariance analysis by gender

A multigroup invariance analysis was carried out with respect to gender for the instrument that was created. Table 10 presents the adjustment values for each model. The results in the table show the absence of differences between the unrestricted model (Model 1) and Model 2 (invariance in the measurement importance). The absence of significant differences between the two models implies a minimum criterion to accept the existence of model invariance by gender of the teaching staff (Byrne et al. 1989; Marsh 1993). Therefore, the UTIC-EEI instrument can be used by both female and male teachers.

3.5 External validity

In relation to gender, the statistical results showed a higher awareness of teachers regarding the use of ICT resources in the dimensions "research" and "teach", compared

Table 7 Factorial correlation matrix (λ = 1)

Factors	**1**	**2**	**3**
1	1.000	.377	.329
2	.377	1.000	.567
3	.329	.567	1.000

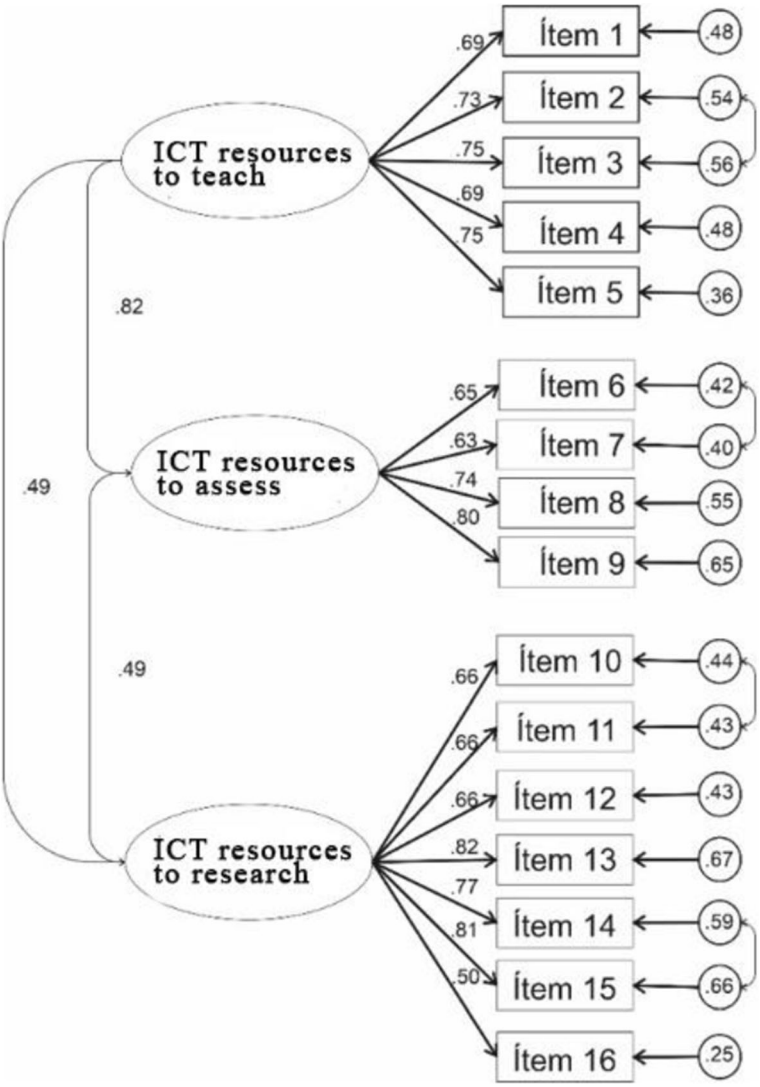


Fig. 2 Structural equation model

to the dimension "evaluate", for both genders. Specifically, no statistically significant differences were found for the gender variable in any of the dimensions of the instrument.

Regarding the variable "type of university", the teaching staff perceived a greater use of ICT resources in the dimensions "research" and "teach" compared to the dimension "evaluate", in both genders. However, in the dimensions of the teaching-learning process (teaching and evaluating), significant differences were found between teachers who work in a private institution and those who work in a public institution, while in the dimension "research", no statistically significant differences were found.

In the labour link variable, the "evaluate" dimension was the one with the least educational use by teachers, for both genders. In addition, the type of connection that

Table 8 Reliability statistics of the three dimensions of the instrument

Dimensions	Coefficients	$N = 447$
DIM. 3	Cronbach's Alpha	.81
	Spearman-Brown coefficient	.77
	Two Guttman halves	.76
DIM. 2	Cronbach's Alpha	.87
	Spearman-Brown coefficient	.77
	Two Guttman halves	.77
DIM. 1	Cronbach's Alpha	.89
	Spearman-Brown coefficient	.85
	Two Guttman halves	.85

teachers have (complete or partial) did not affect the educational use of ICT resources to teach, evaluate, or research.

Finally, no significant differences were found between the four areas of knowledge compared in the dimensions "teach" and "evaluate", thus indicating a similar use for each area of knowledge; however, significant differences were found in the "investigate" dimension Table 11.

4 Discussion and conclusions

As a result of the findings, the instrument designed to evaluate the usage by higher education teachers of ICT resources to teach, evaluate, and carry out research (see Appendix), has proved to be valid and reliable, with a total of 16 items covering all the fundamental dimensions of the professional profile of higher education teachers (Hernández 2009; Cunha 2015; Lazaro 2018).

The theoretical evaluation carried out, which has been defined in an operative manner, which considers the three dimensions of the UTIC-EEI model, was shown to be relevant and unique from other instruments concerning the use of technological resources, since it is rare to find instruments with the characteristics this model comprises, both in terms of the theoretical approach and the operationalisation of its items (Cabero et al. 2010; Correa and Rivadulla-López 2017; Durán et al. 2016; Agreda

Table 9 Correlations between dimensions

	Dimension 1	Dimension 2	Dimension 3
Dimensions 1	1		
Dimensions 2	.639**	1	
Dimensions 3	.414**	.399**	1

** The correlation is significant at the 0.01 level (bilateral)

Table 10 Multigroup Analysis of Invariance by Gender

Models	Invariance by gender								
	χ^2	*gl*	χ^2/gl	$\Delta\chi^2$	Δgl	CFI	IFI	SRMR	RMSEA (IC 90%)
Model 1	350.887	194	1.809	–	–	.954	.955	.060	.043 (.036–.051)
Model 2	365.422	207	1.765	14.54	13	.954	.954	.062	.042 (.035–.049)
Model 3	372.535	213	1.749	7.11*	6	.953	.954	.064	.042 (.035–.049)
Model 4	402.688	233	1.728	25.15**	20	.951	.951	.066	.041 (.034–.048)

*$p < .01$; **$p < .001$

et al. 2017; Tapasco and Giraldo 2017; Taquez et al. 2017; Ge et al. 2018; González et al., 2019; Medina-Rebollo et al. 2019). In addition, the specific educational level to which it is targeted means it is not just one more questionnaire, but rather provides specific and differentiated insight regarding higher education in Spain. This could prove extremely useful for the scientific community, to deepen the knowledge of the use made by university faculty in relation to ICT in teaching, in research, and in the evaluation of their academic activities.

Table 11 Descriptive statistics according to study variables

	M (Sd) Dimensions		
	Teach	Evaluate	Research
Gender			
Female	3.17 (1.07)	2.92 (1.24)	3.81 (0.92)
Male	3.17 (0.98)	2.99 (1.19)	3.87 (0.96)
Sig.	0.975	0.526	0.472
Type of University			
Private	3.62 (0.89)	3.26 (1.09)	3.64 (0.93)
Public	3.10 (1.03)	2.91 (1.22)	3.88 (0.94)
Sig.	0.001*	0.035*	0.078
Connection			
Full	3.12 (1.00)	2.91 (1.19)	3.89 (0.85)
Partial	3.34 (1.10)	3.11 (1.27)	3.69 (1.19)
Sig.	0.67	0.169	0.069
Areas			
Health Sciences	2.88 (1.01)	2.72 (1.14)	3.92 (0.86)
Science and Engineering-Architecture	3.10 (1.05)	3.07 (1.25)	4.07 (0.83)
Science and Engineering-Architecture	3.24 (1.09)	2.83 (1.26)	3.57 (1.11)
Social-Legal Sciences	3.30 (0.97)	3.03 (1.20)	3.78 (0.95)
Sig.	0.477	0.678	0.001*

Grade. *M*: mean, *Sd*: standard deviation,*: $p < .05$.

The validity of the instrument was also fully supported by both the EFA and the CFA, with satisfactory results for each type of analysis and, as a consequence, the correct results of the items in their relevant dimensions, which showed the coherence between the state-of-the-art and the items created. This makes the present study stand out from other studies, which have generally only confirmed content validity through expert judgement (Cabero et al. 2010; Durán et al. 2016; Agreda et al. 2017; Taquez et al. 2017; Rivera-Laylle et al. 2017; Ge et al. 2018; González Jaimes et al. 2019; Medina-Rebollo et al. 2019) and in some cases have also carried out exploratory factor analysis (Agreda et al. 2017; Tapasco and Giraldo 2017; Ge et al. 2018; González Jaimes et al. 2019). Undoubtedly, one of the key elements underlying the performance of both types of construct validity analysis was the size of the sample used in this study (Hair et al. 2009). However, it would have been interesting to study the entire target population in the Spanish territory; this would have increased the generalisation power of the results obtained, since the intentional sampling employed in this study can affect the external validity of research, limiting the generalisability.

Regarding the reliability of the instrument, highly satisfactory psychometric properties were found in each of the dimensions of the questionnaire, and overall (Cronbach's Alpha = .89). Furthermore, the values obtained through this adjustment were also confirmed by other statistics, such as the Spearman–Brown coefficients and Guttman's split-half. In this study, we obtained values higher than those obtained by González Jaimes et al. (2019) and Tapasco and Giraldo (2017).

Furthermore, the results obtained through the analysis of invariance by gender showed that the developed instrument can be administered to both female and male teachers. The previous analysis of instruments used in the theoretical literature did not offer evidence of the psychometric properties of these instruments with respect to this characteristic; this is one of the primary contributions and innovations of the present study in support of the instrument's validity.

In addition, the instrument had external evidence of validity, making it possible to generalise the results generated by this instrument. For example, regarding the gender of the teaching staff, no significant differences were found in the use of ICT resources, in any of the dimensions of the instrument. These findings are evidenced in similar research by other authors that have analysed digital competence with respect to gender (Sevillano-García and Vázquez-Cano 2015; Vázquez-Cano et al. 2017; Dauda et al. 2018). These results justify the use of this questionnaire for both genders, further supporting its validity. Regarding the type of university (public or private), it can be seen that there were no significant differences in relation to the "research" dimension, but there were significant differences with respect to the "teach-evaluate" dimensions. On the other hand, according to the teaching staff's area of knowledge, the only statistically significant differences were found in the research dimension, corroborating the results obtained by Guillén-Gámez et al. (2020), which did not find differences in the other dimensions. Furthermore, no significant differences were found regarding the type of employment relationship (full or partial). The main reason for this situation may be due to the fact that private universities constitute a "company" that must be well positioned in the ranking of universities to attract customers, so

there is a great use of ICT in the teaching-learning processes. In relation to the areas of knowledge, it can be highlighted that the use of ICT can be influenced, to a great extent, by the nature of the area of knowledge itself, since some areas that are more active than others in the field of the investigation.

Ultimately, it can be concluded that the instrument presented here (UTIC-EEI model) is a valid and reliable instrument that can be used by the scientific community in future research, the purpose of which is to increase the use made by higher education teachers of ICT resources, being a key component of the digital competence of said teachers. As a future line of research, it would be beneficial to continue studying the factorial and confirmatory structure of this instrument, in order to establish profiles, in such a way that quality items are selected to analyse each dimension, specifically in the "research" dimension, according to the different areas of knowledge to which the teaching staff belongs, since the use of ICT resources can vary according to this variable. In the same way, it is necessary to study what type of ICT resources are used most frequently in the teaching-learning process in public and private institutions, in order to adapt the questionnaire according to the variable "type of university".

Appendix

Table 12 Self-perception questionnaire regarding the use of ICT resources by university teachers to teach, assess and research. For each of the following items, indicate their degree of use. This is a 5-point Likert scale, where the value 5 is the maximum score to achieve

ICT resources to teach	1	2	3	4	5
1. Learning Management system (LMS) (Blackboard, Moodle, Edmodo)					
2. Collaboration through Google + (Docs, Drive, Sites, meets…					
3. Online tutoring or classes (Hangout, Zoom, Skype)					
4. Content platforms (Blogger, WordPress)					
5. Educational forums to configure activities					
ICT resources to evaluate students	1	2	3	4	5
6. Digital Rubrics through LMS					
7. E-portfolios					
8. Test type controls via LMS					
9. Forums configured to assess activities					
ICT resources for research and publishing scientific articles	1	2	3	4	5
10. Search engines in databases (WoS, Dialnet, Teseo, Eric)					
11. Web search engines to consult bibliography (Google Scholar)					
12. Preparation of references (Refworks, EasyBIB)					
13. Websites of JCR and SJR impact Journals according to their quartiles					
14. Social Networks for research (Researchgate, Academia)					
15. Researcher Profile (Researcher ID, ORCID)					
16. Data analysis software (SPSS, Minitab, Mplus, R, PSPP)					

References

Agreda, M., Hinojo, M.A., y Sola, J.M. (2017). Diseño y validación de un instrumento para evaluar la competencia digital de los docentes en la educación superior española. Pixel-Bit. Revista de Medios y Educación, 49, 39-56. https://doi.org/10.12795/pixelbit.2016.i49.03.

Almeraya, J. M. C., & Tobón, S. (2019). Validez de una rúbrica para medir competencias investigativas en pedagogía desde la socioformación. *Atenas, 3*(47), 1–17 https://atenas.reduniv.edu.cu/index.php/atenas/article/view/486.

Bentler, P. M. (1989). *EQS structural equations program manual*. Los Angeles: BMDP Statistical Software.

Biggs, J. (2005). *Calidad del aprendizaje universitario*. Madrid: Narcea.

Bozu, Z. y Canto, P.J. (2009). El profesorado universitario en la sociedad del conocimiento: competencias profesionales docentes. Revista de Formación e Innovación Educativa en la Universidad, 2 (2), 87–97.

Buela-Casal, G., Guillén-Riquelme, A., Ramiro-Sánchez, T., y Quevedo-Blasco, R. (2017). Ranking de investigación de las universidades públicas españolas. Revista Iberoamericana de Psicología y Salud, 8, 21-35. https://doi.org/10.23923/j.rips.2017.08.003.

Byrne, B. M., Shavelson, R. J., & Muthén, B. (1989). Testing for the equivalence of factor covariance and mean structures: The issue of partial measurement invariance. *Psychological Bulletin, 105*(3), 456–466. https://doi.org/10.1037/0033-2909.105.3.456.

Cabero, J., Marín, V. y Llorente, C. (2010). Hacia el diseño de un instrumento de diagnóstico de competencias tecnológicas del profesorado universitario. Revista Iberoamericana de Educación, 52 (7), 1-12https://doi.org/10.35362/rie5271761.

Cabezas González, M., Casillas Martín, S., Sanches-Ferreira, M., Diogo, T., & Luís, F. (2017). Validación de un instrumento para medir la competencia digital de estudiantes universitarios (CODIEU). Revista de estudios e investigación en Psicología y Educación. (13): 21-24. https://doi.org/10.17979/reipe.2017.0.13.2180.

Calvo, M. B., Abad, F. M., & Conde, M. J. R. (2017). Validación psicométrica de un instrumento de evaluación de competencias informacionales en la educación secundaria. Bordón. *Revista de pedagogía, 69*(1), 27–43. https://doi.org/10.13042/Bordon.2016.48593.

Carretero-Dios, H. y Pérez, C. (2005). Normas para el desarrollo y revisión de estudios instrumentales. International Journal of Clinical and Health Psychology, 5(3), 521–551.

Castro, L., Carvajar, C. y Ugarte, J. (2004). La cultura de la investigación científica en la educación superior en Colombia. Una aproximación desde seis universidades del país. Bogotá. Universidad de los Adres. Progama de Maestría en Dirección Universitaria.

Comisión Europea (2012). Comunicación de la comisión al parlamento europeo, al consejo, al comité económico y social europeo y al comité de las regiones. COM/2018/029 final. Estrasburgo.https://eur-lex.europa.eu/legal-content/BG/TXT/PDF/?uri=CELEX:52018DC0029&from=ES

Correa, M. R., & Rivadulla-López, J. C. (2017). The training of university teachers and students in the use of ICT and Moodle. Educational Contexts: Revista de Educación, (2), 65–81. https://doi.org/10.18172/con.2966.

Cortada de Kohan, N. (1999). *Teorías psicométricas y Construcción de Tests*. Buenos Aires: Editorial Lugar.

Cunha, M. I. (2015). Investigación y docencia: escenarios y senderos epistemológicos para la evaluación de la educación superior. *REDU. Revista de Docencia Universitaria, 13*(1), 79–94.

Dauda, M., Ayanda, M., & Jibrin, A. (2018). Analysis of students' attitude towards the use of information and communication Technology in Nigerian Tertiary Institutions. *ATBU Journal of Science, Technology and Education, 6*(1), 100–105 Available on line at: http://www.atbuftejoste.com/index.php/joste/article/view/487.

Dellit, J. (2001). Using ICT for quality in teaching-learning evaluation processes. Learning federation secretariat Australian education systems officials committee [EN]. http://ictliteracy.info/rf.pdf/UsingICTQuality.pdf

De-Moya-Anegón, F., Herrán-Páez, E., Bustos-González, A., CoreraÁlvarez, E., Tibaná-Herrera, G., y Rivadeneyra, F. (2019). Ranking Iberoamericano de instituciones de educación superior 2019 (SIR Iber). Recuperado de http://www.elprofesionaldelainformacion.com/documentos/SIR_Iber_2019.pdf

Díaz, W. (2016). Formación del profesorado universitario, evaluación de la actividad docente, recursos y promoción profesional. *Estudios Pedagógicos, 42*(1), 65–85. https://doi.org/10.4067/S0718-07052016000100005.

Durán, M., Gutiérrez, I. y Prendes, M. (2016). Certificación de la competencia Tic del profesorado universitario. Diseño y validación de un instrumento. Consejo Mexicano de Investigación Educativa, 21(69), 527–556.

Fan, X., Thompson, B., & Wang, L. (1999). Effects of sample size, estimation methods, and model specification on structural equation modeling fit indexes. *Structural Equation Modeling: A Multidisciplinary Journal, 6*(1), 56–83. https://doi.org/10.1080/10705519909540119.

Fernández-Márquez, E., Leiva-Olivencia, J. J., & López-Meneses, E. (2018). Competencias digitales en docentes de Educación Superior. *Revista Digital de Investigación en Docencia Universitaria, 12*(1), 213–231.

Flores-Lueg, C., & Vila, R. R. (2016). Diseño y validación de una escala de autoevaluación de competencias digitales para estudiantes de pedagogía. Píxel-Bit. Revista de Medios y Educación. (48): 209-224.

From, J. (2017). Pedagogical digital competence–between values, knowledge and skills. *Higher Education Studies, 7*(2), 43–50. https://doi.org/10.5539/hes.v7n2p43.

Ge, W., Han, X., & Shen, X. (2018). Developing a validated instrument to measure Teachers' ICT competencies for university teaching in a digital age. In 2018 seventh international conference of educational innovation through technology (EITT) (pp. 101-105). IEEE. https://doi.org/10.1109/EITT.2018.00028.

Ghomi, M., & Redecker, C. (2018). Digital competence of educators (DigCompEdu): Development and evaluation of a self-assessment instrument for teachers' digital competence. In proceedings of the 11th international conference on computer supported education - volume 1: CSEDU, pages 541–548. https://doi.org/10.5220/0007679005410548.

González Jaimes, E.I., López Chau, A., Trujillo Mora, V. y Bautista López, J. (2019). Instrumento certificador de tecnologías de la información y la comunicación y tecnologías del aprendizaje y el conocimiento para docentes universitarios. Revista Iberoamericana para la investigación y el desarrollo educativo, 10(19), 1-27. Doi: https://doi.org/10.23913/ride.v10i19.516.

Gudmundsdottir, G. B., & Hatlevik, O. E. (2018). Newly qualified teachers' professional digital competence: Implications for teacher education. *European Journal of Teacher Education, 41*(2), 214–231. https://doi.org/10.1080/02619768.2017.1416085.

Guillén-Gámez, F. D., Lugones, A., Mayorga-Fernández, M. J., & Wang, S. (2019). ICT use by pre-service foreign languages teachers according to gender, age and motivation. *Cogent Education, 6*(1), 1574693. https://doi.org/10.1080/2331186X.2019.1574693.

Guillén-Gámez, F. D., Ruiz-Palmero, J., Sánchez-Rivas, E., & Colomo-Magaña, E. (2020). ICT resources for research: An ANOVA analysis on the digital research skills of higher education teachers comparing the areas of knowledge within each gender. Education and information technologies. 1-15. https://doi.org/10.1007/s10639-020-10176-6.

Gutiérrez Castillo, J. J., Cabero Almenara, J., & Estrada Vidal, L. I. (2017). Diseño y validación de un instrumento de evaluación de la competencia digital del estudiante universitario. *Revista Espacios, 38*(10), 1–27 http://revistaespacios.com/a17v38n10/17381018.html.

Hair, J., Anderson, R., Tathan, R. y Black, W. (2009). Análisis multivariante. Madrid: Pearson.

Helleve, I., Grov Almås, A., & Bjørkelo, B. (2019). Becoming a professional digital competent teacher. Professional development in education, 1–13. https://doi.org/10.1080/19415257.2019.1585381.

Henson, R. K., & Roberts, J. K. (2006). Use of exploratory factor analysis in published research: Common errors and some comment on improved practice. *Educational and Psychological Measurement, 66*(3), 393–416. https://doi.org/10.1177/0013164405282485.

Hernández, I. (2009). El docente investigador en la formación de profesionales. *Revista Virtual Universidad Católica del Norte, 27*, 1–21 https://www.redalyc.org/pdf/1942/194215432011.pdf.

Hernández, R. M. (2017). Impacto de las TIC en la educación: Retos y Perspectivas. *Propósitos y representaciones, 5*(1), 325–347.

Hu, L., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure 17 analysis: Conventional criteria versus new alternatives. *Structural Equation 18 Modeling: A Multidisciplinary J, 6*(1), 1–55.

Kelentrić, M., Helland, K., & Arstorp, A. T. (2018). Professional digital competence framework for teachers. *Accessed On, 15*, 1–74.

Kline, R. B. (2005). *Principles and practice of structural equation modeling* (2nd ed.). New York: Guilford.

Lagoa-Varela, D., Álvarez-García, B., y Boedo, L. (2018). Recent changes in the role of Spanish lecturers in economics and business: An empirical analysis based on their own perspectives. Studies in Higher Education, 43, 1321–1333. https://doi.org/10.1080/03075079.2016.1250738.

Lawrence, J. E., & Tar, U. A. (2018). Factors that influence teachers' adoption and integration of ICT in teaching/learning process. *Educational Media International, 55*(1), 79–105. https://doi.org/10.1080/09523987.2018.1439712.

Lazaro, L. M. (2018). Enseñar e investigar hoy en educación comparada: la perspectiva de tres décadas de dedicación académica en ese campo. *RBEC. Revista Brasileña de Educación Comparada, 1*(1), 1–7 https://econtents.bc.unicamp.br/inpec/index.php/rbec/article/view/9387.

Lázaro-Cantabrana, J., Usart-Rodríguez, M., & Gisbert-Cervera, M. (2019). Assessing teacher digital competence: The construction of an instrument for measuring the knowledge of pre-service teachers. *Journal of New Approaches in Educational Research (NAER Journal), 8*(1), 73–78. https://doi.org/10.7821/naer.2019.1.370.

Marsh, H. W. (1993). The multidimensional structure of physical fitness: Invariance over gender and age. *Research Quarterly for Exercise and Sport, 64*(3), 256–273. https://doi.org/10.1080/02701367.1993.10608810.

Martínez-Serrano, M. D. C. D. (2018). Diseño y validación de un cuestionario sobre la competencia digital del alumnado de educación primaria. *Eduser-Revista de Educação, 10*(2), 35–49. https://doi.org/10.34620/eduser.v10i2.113.

Medina-Rebollo, D., Grimaldi-Puyana, M., Del Río-Rama, M., & Sánchez-Oliver, A. (2019). Elaboración y validación de un cuestionario para evaluar el uso didáctico de la plataforma de teleformación Moodle en el contexto universitario. *Revista ESPACIOS, 40*(09), 1–9.

Ministerio de Educación, C. y D. (MECD) (2019). Datos y cifras del sistema universitario español. http://www.educacionyfp.gob.es/dam/jcr:2af709c9-9532-414e-9bad-c390d32998d4/datos-y-cifras-sue-2018-19.pdf

Nordén, L. Å., Mannila, L., & Pears, A. (2017). Development of a self-efficacy scale for digital competences in schools. In 2017 IEEE Frontiers in education conference (FIE) (pp. 1-7). IEEE. https://doi.org/10.1109/FIE.2017.8190673.

Pérez, F., y Aldás, J. (Dirs.) (2018). U-Ranking 2018. Indicadores sintéticos de las universidades españolas (6ª ed.). Madrid: Fundación BBVA. Recuperado de https://www.fbbva.es/wp-content/uploads/2018/06/Informe-U-Ranking-FBBVA-Ivie-2018.pdf.

Rivera-Laylle, L. I., Fernández-Morales, K., Guzmán-Games, F. J. y Eduardo-Pulido, J. (2017). ICT acceptance by university professors: Knowledge, attitude, and practicality. Revista Electrónica Educare, 21(3), 99-116. DOI: https://doi.org/10.15359/ree.21-3.

Salajan, F. D. (2018). An analysis of rhetorical devices in policy narratives on the European Institute of Innovation and Technology: Implications for European higher education. *European Educational Research Journal, 17*, 555–583. https://doi.org/10.1177/1474904117720793.

Sánchez, A., Lagunes, A., Torres, C. A., Judikis, J. C., & López, F. (2019). Exploratory factor analysis of a digital competency questionnaire for research. In Iberoamerican workshop on human-computer interaction (pp. 189-210). Springer, Cham. https://doi.org/10.1007/978-3-030-37386-3_15.

Sánchez-Mena, A., Martí-Parreño, J., & Aldás-Manzano, J. (2018). Teachers' intention to use educational video games: The moderating role of gender and age. Innovations in education and teaching international, 1–12. https://doi.org/10.1080/14703297.2018.1433547.

Sevillano-García, M., & Vázquez-Cano, E. (2015). The impact of digital Mobile devices in higher education. *Journal of Educational Technology & Society, 18*(1), 106–118 Retrieved from http://www.jstor.org/stable/jeductechsoci.18.1.106.

Suwanroj, T., Leekitchwatana, P., & Pimdee, P. (2019). Confirmatory factor analysis of the essential digital competencies for undergraduate students in Thai higher education institutions. *Journal of Technology and Science Education, 9*(3), 340–356. https://doi.org/10.3926/jotse.645.

Tapasco, O. A., & Giraldo, J. A. (2017). Estudio Comparativo sobre Percepción y uso de las TIC entre Profesores de Universidades Públicas y Privadas. *Formación universitaria, 10*(2), 03–12. https://doi.org/10.4067/S0718-50062017000200002.

Taquez, H., Rengifo, D., & Mejía, D. (2017). *Diseño de un instrumento para evaluar el nivel de uso y apropiación de las TIC en una institución de educación superior. Repositorio Digital Universidad Nacional Autónoma de México, 1–30*. Cali, Colombia: Universidad Icesi http://recursos.portaleducoas.org/sites/default/files/5030.pdf.

Thompson, B. (2004). *Exploratory and confirmatory factor analysis: Understanding concepts and applications*. Washington: American Psychological Association.

Tondeur, J., Aesaert, K., Prestridge, S., & Consuegra, E. (2018). A multilevel analysis of what matters in the training of pre-service teacher's ICT competencies. *Computers & Education, 122*, 32–42. https://doi.org/10.1016/j.compedu.2018.03.002.

Tondeur, J., Aesaert, K., Pynoo, B., van Braak, J., Fraeyman, N., & Erstad, O. (2017). Developing a validated instrument to measure preservice teachers' ICT competencies: Meeting the demands of the 21st century. *British Journal of Educational Technology, 48*(2), 462–472. https://doi.org/10.1111.bjet.12380.

Tourón, J., Martín, D., Navarro, E., Pradas, S., & e Iñigo, V. (2018). Construct validation of a questionnaire to measure teachers'digital competence (CDD). *Revista Española de Pedagogía, 76*(269), 25–54. https://doi.org/10.22550/REP76-1-2018-02.

Trejo, R. M. (2017). Diseño Y Validación De Un Instrumento De Evaluación De La Competencia Digital Docente. (Doctoral dissertation, Universitat de les Illes Balears).
Uriel, E., & Aldás, J. (2005). Análisis multivariante aplicado. 1ª. Edición. Thomson. Madrid.
Vázquez-Cano, E., Meneses, E. L., & García-Garzón, E. (2017). Differences in basic digital competences between male and female university students of social sciences in Spain. *International Journal of Educational Technology in Higher Education, 14*(1), 14–27. https://doi.org/10.1186/s41239-017-0065-y.

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